Library Management System

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Our Idea

- Traditional library systems are often outdated and difficult to navigate.
- Students need efficient ways to find, reserve, and manage borrowed books.
- Librarians need efficient tools for inventory management.
- Our project is useful to any library that has not already upgraded to a digital system.

Task Distribution

Ethan: Frontend
Andrew: Backend
Mauricio: Database

Backend

- Research
  - API
  - SQL injection prevention
- File structure created
- Route blueprints created
- Endpoint created to allow librarians to add books
- Endpoint created to filter books
- Started API documentation

Database

Data Base Setup

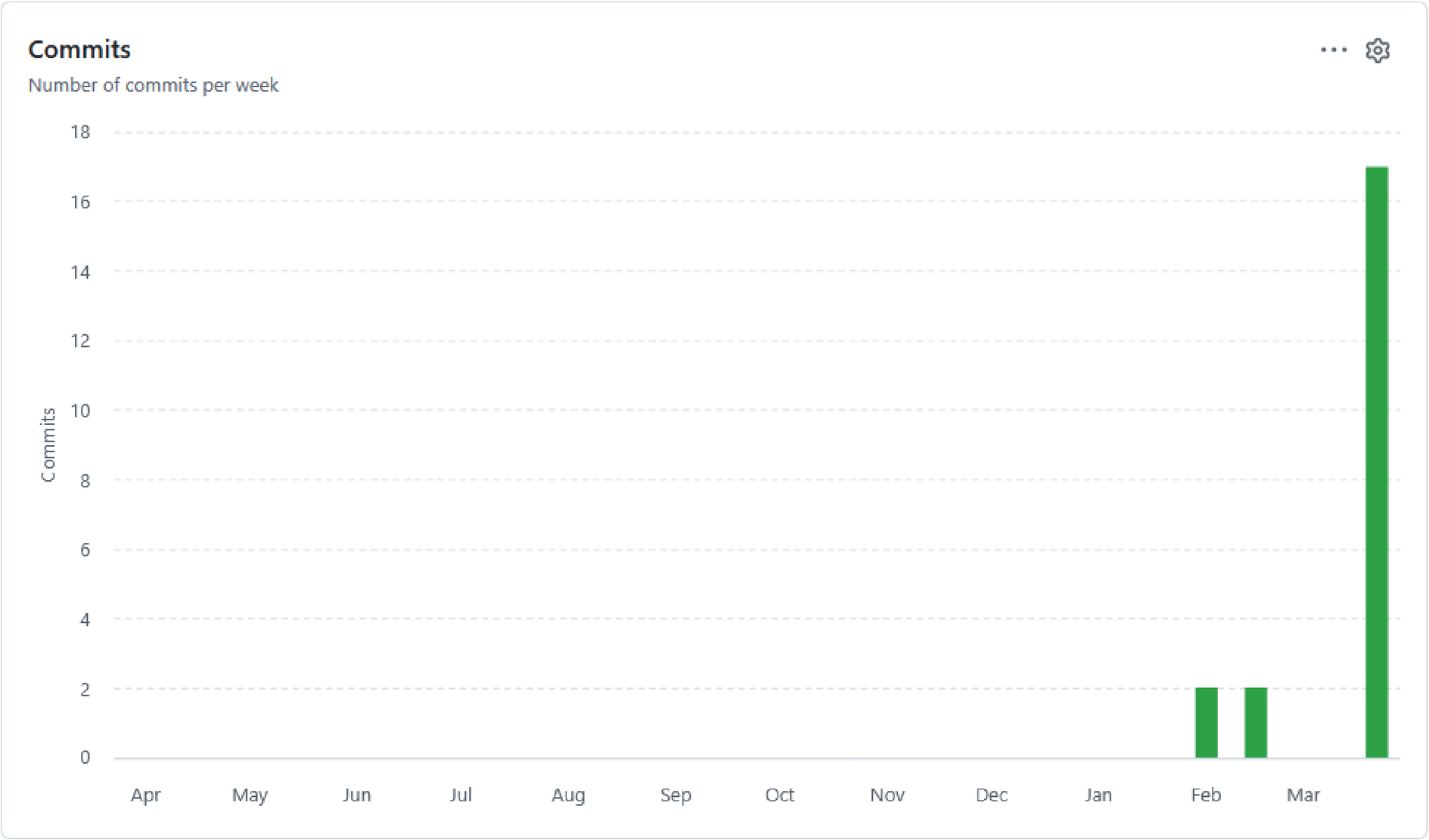
- Built 5 tables: Users, Books, Loans, Fines, and Reservations
- Added a ReadingLists table to support the save favorites feature
- Created an ActiveLoans view for librarians to monitor current checkouts
- Created an UnpaidFines view to track what each user owes
- Created a MostBorrowedBooks view for admin usage reports
- Added 4 sample books to test: checkout → overdue → fine → payment → return

Frontend

- Created website pages for:
- Login
- Dashboard
- Books
- Members
- Loans
- Reservations
- Fines

What’s Left

- Finish the rest of the endpoints for the backend.
- Expand the database
- Test for mobile
- Ensure all inputs are validated
- Improve search and filtering options


Commits
Number of commits per week
Commits
18
16
14
12
10
8
6
4
2
0
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Feb
Mar

Demo